A068 v0.3.0 — PKLSA Intrinsic Phase Clock Actual 3D Euler Dynamics Falsifier

SST-Workbench | Framework v1.0.8

18 October 2026 – development / physical inference unresolved

Abstract

This version *actually evolves* a fully three-dimensional unforced incompressible Euler velocity field initialized from a SHA-256-identified native KnotPlot trefoil, using the unchanged A064 Euler solver, finite-core regularization, Fourier truncation and RK4. Reconstructed 3D vorticity yields an enstrophy-weighted shape coordinate and a separate enstrophy/energy functional; neither has an imposed frequency. Tests cover analytic null controls, conservation, a longer 251-snapshot history, space/time/core refinements and framework provenance. This is a **numerical model** initialized by PKLSA/KnotPlot geometry; it is *not* a confirmed source-native material vortex or an independent physical PKLSA clock. Results below are an **unblinded development precheck**; official BLIND packaging does not retroactively restore epistemic blindness.

1 Identification and certification boundaries

Field	Value
Catalog	A068
Version	v0.3.0
Framework	SST Falsifier Framework v1.0.8 candidate
Profile	`multilibrary_gpu`
Current authority	`EULER_MODEL_SIMULATION_ONLY`
Physical clock	`UNRESOLVED`

2 Research question and objective

Can an autonomous, unforced 3D Euler vortex initialized from native trefoil geometry generate a reproducible intrinsic phase without inserting a clock frequency? What additional independent tests are needed to claim a *material, physical PKLSA* clock? The present version is a numerical prerequisite, not a physical verification.

3 Competing hypotheses and decision semantics

H_0: a numerical transient, slow vortex deformation, rigid-body phase, inadequate sampling, nonconvergent core/mesh effects, or operator-fitting artefact mimics a clock. H_1: a genuine source-native material vortex has a stable relative periodic orbit, coherent intrinsic oscillatory phase and independent withheld operator prediction. An unqualified source produces `UNRESOLVED`, not physical `FAIL`; a failed numerical discretization is not an SST-theory falsification.

4 Assumptions and limiting cases

The 47-point `knot_3.1.txt` geometry belongs to one KnotPlot-native family. Its numerical centerline has no authenticated Euler-equilibrium relaxation, no independently verified source-native topology

and no SI radius. The original A064 centerline is placed in a periodic cubic domain of length $L = 2\pi$ after arbitrary normalization; Gaussian smoothing of its line vorticity uses $a = 0.4$ nominal units and $\Gamma = 1$ model unit. The finite-core filter is a numerical modeling assumption, *not* a derivation of canonical r_c. Actual 3D velocity and vorticity evolve without a driving frequency. A Hilbert-transform phase of a transient need not be a stable clock. Any eventual material-tube or relative periodic orbit qualification requires an additional independent calculation; a passive or moving isosurface is not by itself such proof.

5 Symbols, dimensions and units

Symbol	Meaning	SI dimensions if calibrated
$\mathbf{u}, \boldsymbol{\omega}$	velocity, curl of velocity	LT^{-1}, T^{-1}
L, a, Γ	box length, core width, circulation	L, L, L^2T^{-1}
E	mean energy per unit mass	L^2T^{-2}
H	mean helicity	LT^{-2}
Z	mean enstrophy	T^{-2}
C, λ_i	covariance and its eigenvalues	L^2
q_{sh}, q_Z	two dimensionless functionals	1
$\phi, \widehat{\Omega}$	observed phase, angular slope	$1, T^{-1}$

No dimensional conversion to s, m or kg is made. In particular no SST circulation scale or relativistic time-dilation law is used to set a frequency.

6 Actual continuous and numerical model

The finite-dimensional Galerkin operator implements the projected Lamb form of Euler:

$$\partial_t \mathbf{u} = \mathcal{P}[\mathbf{u} \times \boldsymbol{\omega}], \quad \boldsymbol{\omega} = \nabla \times \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0. \tag{1}$$

The original A064 code deposits a signed line-current along each polygon edge (cloud-in-cell), applies Gaussian finite-core smoothing and a two-thirds wavenumber cutoff, then constructs the divergence-free Fourier Biot–Savart velocity

$$\widehat{\mathbf{u}}_0(\mathbf{k}) = \frac{i\mathbf{k} \times \widehat{\boldsymbol{\omega}}_a(\mathbf{k})}{|\mathbf{k}|^2} \quad (\mathbf{k} \neq 0). \tag{2}$$

RK4 evolves this field without inserted phase. Conservation screens use

$$E = \frac{1}{2V}\int_V |\mathbf{u}|^2 \, dV, \quad H = \frac{1}{V}\int_V \mathbf{u} \cdot \boldsymbol{\omega} \, dV, \quad Z = \frac{1}{2V}\int_V |\boldsymbol{\omega}|^2 \, dV. \tag{3}$$

For the full periodic volume, use enstrophy weights to construct the periodic-centered three-dimensional covariance tensor C. Ordered eigenvalues define the shape scalar

$$q_{\mathrm{sh}} = \frac{\lambda_1 - \lambda_2}{\operatorname{tr} C}, \quad q_{\mathrm{sh},2} = \frac{\lambda_2 - \lambda_3}{\operatorname{tr} C}. \tag{4}$$

These two eigenvalue gaps are correlated. Instead of treating them as independent physical clocks, use the distinct flow functional

$$q_Z = \frac{L^2 Z}{E}. \tag{5}$$

Both q_{sh} and q_Z are computed from each actual evolved 3D field, not from imposed oscillators. The analytic-signal phase estimator is

$$\phi_q(t) = \operatorname{unwrap} \arg\{[q(t) - \bar{q}] + i\mathcal{H}[q(t) - \bar{q}]\}, \qquad \widehat{\Omega}_q = \operatorname{slope}_W \phi_q(t). \tag{6}$$

This estimator is a screen only; it does not establish a relative periodic orbit or intrinsic clock. Thresholds were specified in method preflight: at least six unwrapped cycles; RMS phase-fit error at most 0.20 radians; instantaneous frequency CV at most 0.12; cross-frequency relative error at most 0.03. A constant shape under rigid rotation is no clock. A few cycles of a windowed transient are not a proof of stability. Time series are uniformly sampled by construction.

Machine-readable science contract

Research question Does an original KnotPlot 3_1 centreline, used only as finite-core Euler initial data, generate a converged internally recurring phase under autonomous 3D incompressible Euler evolution, and under what additional criteria may this become a physical PKLSA clock?

Objective Compute actual time-resolved 3D Euler velocity/vorticity from source-native trefoil XYZ with no imposed frequency, evaluate shape and enstrophy phase, conservation, spatial/time/core convergence, and keep physical PKLSA gate G3 unresolved without material confinement and independent operator/provider qualification.

H_0 H0: current source-initialized Euler dynamics provides no qualified, reproducible material-vortex intrinsic clock; apparent Hilbert phase may be transient deformation, core/mesh artefact or spectral leakage.

H_1 H1: after independent material tube, relative periodic orbit and source qualification, the Euler/PKLSA vortex has a coherent recurrent phase stable under mesh, timestep and finite-core changes with a withheld Floquet frequency prediction.

Registered equations

E1: Unforced incompressible 3D Euler in projected Lamb form

$$\partial_t \mathbf{u} = \mathcal{P}[\mathbf{u} \times (\nabla \times \mathbf{u})], \qquad \nabla \cdot \mathbf{u} = 0 \tag{7}$$

Dimensional check: Both sides L T^-2; projection is dimensionless.

E2: Biot–Savart initialization from source-native line vorticity

$$\widehat{\mathbf{u}}(\mathbf{k}, 0) = \frac{i\mathbf{k} \times \widehat{\boldsymbol{\omega}}_a(\mathbf{k})}{|\mathbf{k}|^2}, \qquad \mathbf{k} \neq 0 \tag{8}$$

Dimensional check: i k cross omega_hat / k^2 gives velocity dimension L T^-1.

E3: Integral diagnostics energy, helicity, enstrophy

$$E = \frac{1}{2V}\int_V |\mathbf{u}|^2 dV, \quad H = \frac{1}{V}\int_V \mathbf{u} \cdot \boldsymbol{\omega}\, dV, \quad Z = \frac{1}{2V}\int_V |\boldsymbol{\omega}|^2 dV \tag{9}$$

Dimensional check: E: L^2 T^-2; H: L T^-2; Z: T^-2.

E4: Rotation-invariant shape observable from full 3D vorticity

$$q_{\mathrm{sh}}(t) = \frac{\lambda_1[C(t)] - \lambda_2[C(t)]}{\operatorname{tr} C(t)} \tag{10}$$

Dimensional check: Eigenvalue difference and covariance trace both L^2.

E5: Distinct dimensionless scalar flow functional

$$q_Z(t) = \frac{L^2 Z(t)}{E(t)} \tag{11}$$

Dimensional check: L^2 times T^-2 divided by L^2 T^-2 = 1.

E6: Signal-derived Hilbert phase and frequency without injection

$$\phi_q(t) = \operatorname{unwrap} \arg(q - \overline{q} + i\mathcal{H}[q - \overline{q}]), \qquad \widehat{\Omega}_q = \operatorname{slope}_{t \in W} \phi_q(t) \tag{12}$$

Dimensional check: q and phase dimensionless, Omega T^-1.

E7: Noncommuting temporal, spatial and core convergence probes

$$\delta_Z(A, B) = \frac{|Z_A(T) - Z_B(T)|}{\max(|Z_B(T)|, \epsilon_Z)} \tag{13}$$

Dimensional check: Ratio of two enstrophy magnitudes dimensionless; epsilon_Z must have same units as Z.

Registered code-to-mathematics steps

1. **S1** [G1; equations E2]: Hash 6 source files and exact A064 solver copy, reject mutations
2. **S2** [G1; equations E6]: Unblind method controls positive oscillator, rigid rotation and beating; preserve as method-only
3. **S3** [G2; equations E2]: Normalize unchanged native XYZ into periodic 2pi cube; deposit signed circulation; apply Gaussian finite-core and transverse Leray projection
4. **S4** [G2; equations E1, E3]: Run genuine unforced spectral 3D Euler/RK4 trajectory to T=20, n16 dt.04 with 251 snapshots; compute energy/helicity/CFL/divergence
5. **S5** [G3; equations E4, E5, E6]: Compute independent functionals qshape and qZ from evolved 3D fields then Hilbert phase, no oscillator injected
6. **S6** [G5; equations E1, E7]: Rerun n24, n16 dt=.02 and two finite-core branches (.32,.48) to fixed T, report differences even when science gate G5 is blocked
7. **S7** [G3; equations E4, E6]: Do not mark G3 terminal without qualified material tube and relative periodic orbit; do not turn unresolved into fail
8. **S8** [G4; equations E1, E6]: Register future independent Euler/Floquet frequency on a withheld provider before running G4-G8

7 Executable algorithm and fixed run matrix

1. Verify byte-exact historical controls and A064 solver/source hashes; refuse changed or missing sources.
2. Load 47 native 3D XYZ vertices; normalize and map to the periodic box; deposit vorticity and compute solenoidal $\mathbf{u}(0)$ from A064.
3. In `BASIC`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 6$, retaining 151 genuine field snapshots.

4. In `FULL/CERTIFY`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 20$, retaining 251 3D fields. Independently evolve $N = 24$ at identical $(a, \Delta t, T)$, $N = 16$ at $\Delta t = 0.02$, and $N = 16$ with $a = 0.32, 0.48$ at fixed $(T, \Delta t)$.

5. At every retained field, compute E, H, Z, divergence, CFL, shape tensor, $q_{\rm sh}$ and q_Z. Recover phase only from measured functionals; do not input a theoretical frequency.

6. Require numerical conservation to close G2; record spatial/time/core diagnostic differences even when downstream physical gates are blocked. Leave G3 unresolved without a material tube, a verified relative periodic orbit, independent providers and independent operator frequency prediction.

7. Let the official v1.0.8 runtime write unique BLIND ZIPs, PDFs and gate-ledger hashes; do not bypass reveal policy.

8 Source provenance and provider independence

`knot_3.1.txt`: SHA-256 `98a628af4ba5d09274f0a1845c351a7a3d633a39ff833a3ae0b78e351bafdabf`. The original A064 Python Euler solver is included byte-for-byte at `experiment/euler3d.py` and as historical reference. Earlier two A068 preflight JSONs and A064-v0.3.0 results are immutable controls, not independent vortex providers. The finite-core smoothing function, Fourier truncation and normalization are inherited from A064. The expected upstream hashes are enforced in G1, and mesh/core variation does not create distinct provider families. Input XYZ geometry has *unverified* physical vortex equilibrium and no SI units.

9 Dependency graph and physical guardrails

Gate	Required decision	Initial status
G0	Framework freeze, signed provenance path	runtime
G1	Native geometry, solver and historical control hashes	source check
G2	Real Euler 3D evolution plus $E, H, \nabla \cdot \mathbf{u}$, CFL	numerical PASS/FAIL
G3	Six clean cycles; *material tube and RPO independently qualified*	UNRESOLVED
G4	Independent withheld Euler/Floquet prediction	blocked
G5	Converged space, time and core with physical state fixed	blocked
G6	Explicit native backend parity/authority	blocked
G7	Multiple independent qualified PKLSA providers	blocked
G8	Physical mechanism conclusion if all required gates pass	blocked
G9	Nonced reveal commitments on allowed frozen gate ledger	withheld

Gate order is an explicit DAG. A negative numerical screen does not count as a physical hypothesis falsification. A gapped clock phase without an identified finite-core material vortex can *never* close G3 in this release.

10 Numerical control and exploratory observations

For G2 preregistered development thresholds: relative divergence $< 10^{-9}$; relative energy drift $< 10^{-5}$; relative helicity drift $< 10^{-4}$; CFL < 0.25; finite velocity field, and nontrivial evolution. Comparative tests use

$$\delta_Z(A, B) = \frac{|Z_A(T) - Z_B(T)|}{\max(|Z_B(T)|, 10^{-30})}. \tag{14}$$

Exploratory, **unblinded** local FULL prechecks (not an official frozen framework run):

Case	Grid, core, Δt	$Z(T)$	$\lvert\Delta E/E\rvert_{\max}$	Phase cycles
Base	$16, 0.40, 0.04$	0.064202	8.34×10^{-10}	0.646
Spatial	$24, 0.40, 0.04$	0.121697	6.93×10^{-9}	0.615
Temporal	$16, 0.40, 0.02$	0.064202	4.22×10^{-11}	0.646
Core low	$16, 0.32, 0.04$	0.094750	1.69×10^{-9}	0.643
Core high	$16, 0.48, 0.04$	0.035642	2.04×10^{-10}	0.723

δ_Z under time-step halving is 2.01×10^{-9}, while spatial $\delta_Z \simeq 0.472$ and core-spread $\simeq 0.921$. Thus temporal integration control succeeds but spatial/core physical convergence demonstrably **does not** qualify. Less than one observed phase cycle fails the method screen's six-cycle minimum; it cannot be extrapolated to a period by assumption. These values are preliminary numerical model diagnostics and will be recomputed on each framework run.

11 Backend authority

Actual field integration is the upstream NumPy/SciPy FP64 Euler simulator. Its runs are not equivalent to native C++ FP64 or Intel SYCL parity certification. Framework v1.0.8 backend configuration may disable certain gates; disabled scientific gates do not pass. DD32/FP32 only screen, never physical confirmation. All core/mesh branches here share one upstream mathematical method and source provenance.

12 Blindness, frozen protocol and reveal

The historical source and this exploratory run have already been inspected, so only framework *output confidentiality and audit integrity* are blind. Future provider-independent PKLSA-clock claims must be preregistered on a fresh held-out source before opening it. Frozen v1.0.8 publishes run-specific ZIP/PDF identities and archives previous output roots; `REVEAL_IF_ALLOWED` correctly refuses when a required physical G8 is `NOT_RUN_PREREQUISITE`. Never edit a frozen policy, ledger or HMAC commitment to force a reveal.

13 Statistics and confidence limits

The Euler trajectory is one deterministic single-provider numerical realization. Time snapshots, mesh steps and smoothing variants are *not* independent experimental replicates. A Hilbert phase slope on less than one cycle has no established asymptotic period. No physical confidence interval or p-value is computed. Discovery windows and candidate selection are fixed by code; no frequency target, SST phase law or operator eigenvalue is supplied.

14 Actual runtime results and retained evidence

Each official BASIC/FULL/CERTIFY pipeline computes and stores `EULER_DYNAMICS_RESULTS.json`, `EULER_3D_FIELDS_MODEL_ONLY.npz` with $t, \mathbf{u}, \mathbf{q}$ and conservation tracks, the hashed source manifest, full gate ledger and framework result PDF. The NPZ includes actual evolved velocity on the 3D grid, not just 2D projections or A064 detector curves. Only the *primary* trajectory is retained as full fields; full diagnostics from all refinement branches are in JSON.

Automatic run provenance

Field	Recorded value
Run ID	20261018T011646784165Z-2ac8031a

Run phase	BLIND
Protocol bundle SHA-256	7399a9e78cb9f5ee45844bc8481229496fe8de1bb96ddbfc2526373f16936737
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	ed18a4bd43126b396934cad8c3b5408fe370c1c39c1c97a6076325313baefdaf

Gate ledger

Gate	Status	Question
G0	PASS	Did frozen framework v1.0.8 verify this versioned scientific protocol?
G1	PASS	Do native KnotPlot trefoil, A064 Euler source and earlier preflight ha
G2	PASS	Does the geometry-seeded 3D Fourier-Galerkin Euler trajectory pass c
G3	UNRESOLVED	Does a registered, materially bounded PKLSA vortex have an intrinsi
G4	NOT_RUN_PREREQUISITE	Does a withheld autonomous Euler linearization/Floquet frequency pr
G5	NOT_RUN_PREREQUISITE	Do spatial, temporal and finite-core refinements converge for the same
G6	NOT_RUN_PREREQUISITE	Does independent CPU FP64 / SYCL screening meet backend parity
G7	NOT_RUN_PREREQUISITE	Are at least two genuinely independent PKLSA dynamics providers co
G8	NOT_RUN_PREREQUISITE	Is the final physical mechanism eligible for a decisive PASS or FAIL a
G9	DEFERRED	Reveal verification after registered blind physical conclusion.

Backend manifest


```
{
  "actual_backend": "python-numpy-fp64",
  "authority": "EULER_MODEL_SIMULATION_ONLY",
  "full_3d_trajectory_file": "EULER_3D_FIELDS_MODEL_ONLY.npz",
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  "operator_floquet_prediction": "NOT_EXECUTED",
  "physical_intrinsic_clock": "UNRESOLVED",
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    "max_cfl": 0.0581718652009979,
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    "max_relative_divergence": 8.787215805350529e-16,
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  "refinement_convergence": {
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    "finite_core_robust_gate": false,
    "spatial_gate": false,
```

```
    "spatial_relative_Z": 0.472447314642971,
    "temporal_gate": true,
    "temporal_relative_Z": 2.0135566771611314e-09
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "simulation_model": "PERIODIC_DEALIASED_3D_EULER_RK4",
  "source_hash_ok": true,
  "version": "A068-v0.3.0"
}
```


Source manifest


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    "hits": []
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  "role": "seed_geometry",
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  "version": "v0.3.0"
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  "version": "v0.3.0"
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```

```
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      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation and next falsification step

The existence of nonlinear Euler evolution does not demonstrate a stable toroidal material boundary, spectral convergence, global RPO or an intrinsic physical clock. Future v0.4 qualification needs a source-native independent core field, advected material boundary with demonstrated low flux, full-state recurrence modulo allowed symmetries, a separately computed time-dependent Floquet/Koopman prediction and a new source/protocol commitment. Historical A064 spectral eigenpairs at $N = 8, 12, 16$ were nonconvergent and cannot be used as the predicted clock frequency here. Likewise, no comparison to Lorentz time dilation is permitted before a physical intrinsic oscillator is certified.

16 Reproducibility and bibliography

Install only into a *new* A068-v0.3.0 instance with the official `tools/new_falsifier.py` from locally present v1.0.8. Run `pytest -q tests`, then `run_all.cmd FREEZE`, `run_all.cmd BASIC`, `run_all.cmd FULL`, and optionally `run_all.cmd CERTIFY`; no frozen package from v0.2.0 is overwritten. Preserve all v0.2.0 result evidence and commits.

References

[1] P. G. Saffman, *Vortex Dynamics* (Cambridge University Press, 1992), doi:10.1017/CBO9780511624063.

[2] H. K. Moffatt, J. Fluid Mech. **35**, 117–129 (1969), doi:10.1017/S0022112069000991.

[3] D. Gabor, J. Inst. Elect. Eng. Part III **93**, 429–457 (1946), doi:10.1049/ji-3-2.1946.0074.